

Addendum

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TCGA-BJ-A45H-01A-PR

Redacted

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block C (Thyroid, Right):

- A. **BRAF** codon 600 mutation IDENTIFIED (p.V600E, c.1799T>A).
- B. Mutations in **NRAS61**, **HRAS61**, **KRAS12/13** and **RET/PTC1** and **RET/PTC3** rearrangements NOT Identified.

Block F (Thyroid, Left):

- A. **BRAF** codon 600 mutation IDENTIFIED (p.V600E, c.1799T>A).
- B. Mutations in **NRAS61**, **HRAS61**, **KRAS12/13** and **RET/PTC1** and **RET/PTC3** rearrangements NOT Identified.

NOTE:

Nucleic acids were extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either **BRAF** or **RAS** genes or **RET/PTC** rearrangements are found in more than 70% of papillary thyroid carcinomas (1). Mutations in the **RAS** genes or **PAX8/PPAR γ** rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (2). Finding of **BRAF**, **RET/PTC** or **PAX8/PPAR γ** mutations in thyroid FNA samples correlates with ~100% probability of cancer and finding of **RAS** mutation correlates with malignancy in 74-85% of cases (3-5). Based on our prospective study at risk of cancer in thyroid nodules negative for all of these mutations is 5.9% in nodules with atypia of undetermined significance/follicular lesion of undetermined significance (AUS/FLUS) cytology (including the 2.3% risk of invasive cancer and 0.5% risk of cancer with extrathyroidal spread in this group), 14% in nodules with follicular neoplasm/suspicious for follicular neoplasm (FN/SFN) cytology, and 28% in nodules with suspicious for malignant cells cytology (5). In addition, **BRAF** V600E (T1733A) mutation has been associated with more aggressive behavior of papillary carcinoma (6) and may be used for risk stratification of patients with papillary thyroid cancer (7). However, single cases of false-positive **BRAF** mutation detection in thyroid FNA samples have been reported in the literature (8); therefore, the results of molecular testing should be interpreted in combination with clinical information, imaging, and cytology.

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006; 30:216-222
2. Nikiforova MN, et al. RAS point mutations and PAX8-PPAR γ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003; 88: 2318-26.
3. Nikiforov YE et al. Molecular testing for mutations in improving the fine-needle aspiration diagnosis of thyroid nodules. *J Clin Endocrinol Metab*. 2009 Jun;94(6):2092-8.
4. Cantare S, et al. Impact of proto-oncogene mutation detection in cytological specimens from thyroid nodules improves the diagnostic accuracy of cytology. *J Clin Endocrinol Metab* 2010; 95:1365-1369.
5. Nikiforov YE, et al. Impact of Mutational Testing on the Diagnosis and Management of Patients with Cytologically Indeterminate Thyroid Nodules: A Prospective Analysis of 1056 FNA Samples. *J Clin Endocrinol Metab*, 2011 98(11):3390-7.
6. Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008; 93:3943-3949.
7. Xing M, et al. BRAF mutation testing of thyroid fine-needle aspiration biopsy specimens for preoperative risk stratification in papillary thyroid cancer. *J Clin Oncology* 2010; 27:2977-82.
8. Chung KW, et al. Detection of BRAFV600E mutation on fine needle aspiration specimens of thyroid nodule refines cyto-pathology diagnosis, especially in BRAF600E mutation-prevalent area. *Clin Endocrinol (Oxf)*. 2008; 65(5):666-8.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 80% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the LightCycler platform to amplify **BRAF** codons 599-601, **NRAS** codon 61, **HRAS** codon 61, and **KRAS** codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of **RET/PTC1** and **RET/PTC3** rearrangements on ABI7500. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on ABI3130. Amplification of the **KRT7** genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of cells with mutation present in the background of normal DNA and RNA.

FINAL DIAGNOSIS:

Collection Date:

THYROID, TOTAL THYROIDECTOMY (20 g) -

- A. **PAPILLARY THYROID CARCINOMA, TWO FOCI, BILATERAL (RIGHT: 1.5 cm, LEFT: 0.5 cm) WITH EXTRA THYROIDAL EXTENSION AND POSITIVE MARGINS ON LEFT SIDE.**
- B. **BOTH FOCI: CONVENTIONAL TYPE**
- C. **ANGIOLYMPHATIC INVASION PRESENT.**
- D. **PATHOLOGIC STAGE: pT3NX**

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:

TUMOR SITE:

TUMOR FOCALITY:

TUMOR SIZE (largest nodule):

HISTOLOGIC TYPE**:

PRIMARY TUMOR (pT):

REGIONAL LYMPH NODES (pN):

DISTANT METASTASIS (pM):

EXTRATHYROIDAL EXTENSION:

MARGINS:

LYMPH-VASCULAR INVASION:

Total Thyroidectomy
Right Lobe, Left Lobe

Multifocal

Greatest Dimension: 1.5 cm

Papillary carcinoma

pT3

pNX

Not applicable

Present

Margin(s) involved by carcinoma

Site(s) of involvement: Left Lobe

Present

Criteria	Yes	No
Diagnosis Discrepancy		☒
Primary Tumor Site Discrepancy	☒	
HPAA Discrepancy		☒
Prior Malignancy History		☒
Dual/Synchronous Primary Noted		☒
Case is (circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	BTH	Date Reviewed: 8/26/2012

ICD-O-3
carcinoma, papillary, thyroid 8260/3
Site: thyroid, nos C73.9
hw 9/25/12